L15 series solution

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Engineering Mathematics III

ENGE702

Lecture 15: Series solutions

Power series

Recall that a *power series* is an expression such as

$$\sum_{k=0}^{\infty} a_k (x-x_0)^k = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + a_3(x-x_0)^3 + \cdots,$$

where $a_0, a_1, a_2, \ldots, a_k, \ldots$ and x_0 are constants.

Most functions can be written in terms of a power series such as

$$e^x = \sum_{k=0}^{\infty} \frac{1}{k!} x^k.$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}.$$

Power series of $f(x)$:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n.$$

Example: power series of e^x at $x_0 = 0$.

$f(x) = e^x \quad f(0) = e^0 = 1$

$f'(x) = e^x \quad f'(0) = e^0 = 1$

$f''(x) = e^x \quad f''(0) = e^0 = 1$

$\vdots$

$f^{(n)}(x) = e^x \quad f^{(n)}(0) = e^0 = 1$

So at $x_0 = 0$, $e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x)^n = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$.

Series solutions of linear second order ODEs

Suppose we have a linear second order homogeneous ODE of the form

$$P(x)y'' + Q(x)y' + R(x)y = 0.$$

Suppose that $P(x)$, $Q(x)$, and $R(x)$ are polynomials. We will try a solution of the form

$$y = \sum_{k=0}^{\infty} a_k (x-x_0)^k$$

and solve for the a_k to try to obtain a solution defined in some interval around x_0.

The point x_0 is called an *ordinary point* if $P(x_0) \neq 0$. That is, the functions

$$\frac{Q(x)}{P(x)} \quad \text{and} \quad \frac{R(x)}{P(x)}$$

are defined for x near x_0. If $P(x_0) = 0$, then we say x_0 is a *singular point*. Handling singular points is harder than ordinary points and so we now focus only on ordinary points.

For a 2nd ODE with polynomial coefficients $P(x)$, $Q(x)$ and $R(x)$,

$P(x) y'' + Q(x) y' + R(x) y = 0$,

we want to find $y(x)$ as a power series form:

$$y(x) = \sum_{k=0}^{\infty} a_k (x-x_0)^k.$$

We start with a very simple example

$$y'' - y = 0.$$

Let us try a power series solution near $x_0 = 0$, which is an ordinary point. Every point is an ordinary point in fact, as the equation is constant coefficient.

We try

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

If we differentiate, the $k = 0$ term is a constant and hence disappears. We get

$$y' = \sum_{k=1}^{\infty} k a_k x^{k-1}.$$

We differentiate yet again to obtain (now the $k = 1$ term disappears)

$$y'' = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2}.$$

Example: Solve $y'' - y = 0$ for a series solution $y(x) = \sum_{k=0}^{\infty} a_k (x-x_0)^k$.

$P(x) y'' + Q(x) y' + R(x) y = 0$

So $P(x) = 1$, $Q(x) = 0$, $R(x) = -1$ in this example.

At $x_0 = 0$, then $P(x_0) = 1 \neq 0$ so it is ordinary point.

For the series solution $y(x) = \sum_{k=0}^{\infty} a_k x^k$.

We reindex the series (replace k with $k+2$) to obtain

$$y'' = \sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k.$$

We reindex the series (replace k with $k+2$) to obtain

$$y'' = \sum_{k=0}^{\infty} (k+2)(k+1)\, a_{k+2} x^k.$$

Now we plug y and y'' into the differential equation

$$\begin{aligned} 0 = y'' - y &= \left(\sum_{k=0}^{\infty} (k+2)(k+1)\, a_{k+2} x^k \right) - \left(\sum_{k=0}^{\infty} a_k x^k \right) \\ &= \sum_{k=0}^{\infty} \Big((k+2)(k+1)\, a_{k+2} x^k - a_k x^k \Big) \\ &= \sum_{k=0}^{\infty} \big((k+2)(k+1)\, a_{k+2} - a_k \big)\, x^k. \end{aligned}$$

As $y'' - y$ is supposed to be equal to 0, we know that the coefficients of the resulting series must be equal to 0. Therefore,

$$(k+2)(k+1)\, a_{k+2} - a_k = 0, \qquad \text{or} \qquad a_{k+2} = \frac{a_k}{(k+2)(k+1)}.$$

The equation above is called a *recurrence relation* for the coefficients of the power series. It did not matter what a_0 or a_1 was. They can be arbitrary. But once we pick a_0 and a_1, all other coefficients are determined by the recurrence relation.
Let us see what the coefficients must be. First, a_0 and a_1 are arbitrary. Then,

$$a_2 = \frac{a_0}{2}, \quad a_3 = \frac{a_1}{(3)(2)}, \quad a_4 = \frac{a_2}{(4)(3)} = \frac{a_0}{(4)(3)(2)}, \quad a_5 = \frac{a_3}{(5)(4)} = \frac{a_1}{(5)(4)(3)(2)}, \quad \cdots$$

So for even k, that is $k = 2n$, we have

$$a_k = a_{2n} = \frac{a_0}{(2n)!},$$

and for odd k, that is $k = 2n+1$, we have

$$a_k = a_{2n+1} = \frac{a_1}{(2n+1)!}.$$

Let us write down the series

$$y = \sum_{k=0}^{\infty} a_k x^k = a_0 \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n} + a_1 \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1}.$$

Let us solve a more complex example. Consider *Airy's equation*[1]:

$$y'' - xy = 0,$$

near the point $x_0 = 0$. Note that $x_0 = 0$ is an ordinary point.
We try

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

We differentiate twice (as above) to obtain

$$y'' = \sum_{k=2}^{\infty} k(k-1)\, a_k x^{k-2}.$$

We plug y into the equation

$$\begin{aligned} 0 = y'' - xy &= \left(\sum_{k=2}^{\infty} k(k-1)\, a_k x^{k-2} \right) - x \left(\sum_{k=0}^{\infty} a_k x^k \right) \\ &= \left(\sum_{k=2}^{\infty} k(k-1)\, a_k x^{k-2} \right) - \left(\sum_{k=0}^{\infty} a_k x^{k+1} \right). \end{aligned}$$

[1] Named after the English mathematician Sir George Biddell Airy (1801–1892).

We reindex to make things easier to sum

$$\begin{aligned} 0 = y'' - xy &= \left(2a_2 + \sum_{k=1}^{\infty} (k+2)(k+1)\, a_{k+2} x^k \right) - \left(\sum_{k=1}^{\infty} a_{k-1} x^k \right) \\ &= 2a_2 + \sum_{k=1}^{\infty} \big((k+2)(k+1)\, a_{k+2} - a_{k-1} \big)\, x^k. \end{aligned}$$

Again $y'' - xy$ is supposed to be 0, so $a_2 = 0$, and

$$(k+2)(k+1)\, a_{k+2} - a_{k-1} = 0, \qquad \text{or} \qquad a_{k+2} = \frac{a_{k-1}}{(k+2)(k+1)}.$$

We jump in steps of three. First, since $a_2 = 0$, we must have $a_5 = 0$, $a_8 = 0$, $a_{11} = 0$, etc. In general, $a_{3n+2} = 0$.
The constants a_0 and a_1 are arbitrary and we obtain

$$a_3 = \frac{a_0}{(3)(2)}, \quad a_4 = \frac{a_1}{(4)(3)}, \quad a_6 = \frac{a_3}{(6)(5)} = \frac{a_0}{(6)(5)(3)(2)}, \quad a_7 = \frac{a_4}{(7)(6)} = \frac{a_1}{(7)(6)(4)(3)}, \quad \cdots$$

For a_k where k is a multiple of 3, that is, $k = 3n$, we notice that

$$a_{3n} = \frac{a_0}{(2)(3)(5)(6)\cdots(3n-1)(3n)}.$$

For a_k where $k = 3n+1$, we notice

$$a_{3n+1} = \frac{a_1}{(3)(4)(6)(7)\cdots(3n)(3n+1)}.$$

In other words, if we write down the series for y, it has two parts

$$y = \left(a_0 + \frac{a_0}{6} x^3 + \frac{a_0}{180} x^6 + \cdots + \frac{a_0}{(2)(3)(5)(6)\cdots(3n-1)(3n)} x^{3n} + \cdots \right)$$

At $x_0 = 0$, then $P(x_0) = 1$ so it is ordinary point.

For the series solution $y(x) = \sum_{k=0}^{\infty} a_k x^k$.

$\Rightarrow y'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}$.

$\Rightarrow y''(x) = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2}$

reindex series by replacing k with $k+2$ to get

$$y''(x) = \sum_{k=0}^{\infty} (k+2)(k+2-1)\, a_{k+2}\, x^{(k+2)-2}$$
$$= \sum_{k=0}^{\infty} (k+2)(k+1)\, a_{k+2}\, x^k.$$

Then the ODE becomes:

$$y'' - y = \sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} \boxed{x^k} - \sum_{k=0}^{\infty} a_k \boxed{x^k} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} \left[(k+2)(k+1) a_{k+2} - a_k \right] x^k = 0$$

Therefore $(k+2)(k+1) a_{k+2} - a_k = 0$

$\Rightarrow a_{k+2} = \dfrac{a_k}{(k+2)(k+1)}$ recurrence relation.

$a_2 = \dfrac{a_0}{2 \times 1}$ $\quad a_4 = \dfrac{a_2}{4 \times 3} = \dfrac{a_0}{4 \times 3 \times 2 \times 1}$

$a_6 = \dfrac{a_4}{6 \times 5} = \dfrac{a_0}{6 \times 5 \times 4 \times 3 \times 2 \times 1} = \dfrac{a_0}{6!}$

$a_3 = \dfrac{a_1}{3 \times 2}$ $\quad a_5 = \dfrac{a_3}{5 \times 4} = \dfrac{a_1}{5 \times 4 \times 3 \times 2} = \dfrac{a_1}{5!}$

For even $k = 2n$ $\quad a_k = \dfrac{a_0}{k!} = \dfrac{a_0}{(2n)!}$

For odd $k = 2n+1$ $\quad a_k = \dfrac{a_1}{k!} = \dfrac{a_1}{(2n+1)!}$

So the series solution:

$$y = \sum_{k=0}^{\infty} a_k x^k = a_0 \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n} + a_1 \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1}$$

Airy equation $y'' - xy = 0$.

$P(x) = 1$, $Q(x) = 0$, $R(x) = -x$.

At $x_0 = 0$, $P(x_0) = 1 \neq 0$, ordinary point.

For $y = \sum_{k=0}^{\infty} a_k x^k$, $y' = \sum_{k=1}^{\infty} k a_k x^{k-1}$, $y'' = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2}$

reindex, replace k with $k+2$, so $y'' = \sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k$.

reindex $xy = \sum_{k=0}^{\infty} a_k x^{k+1}$, replace k with $k-1$

$$= \sum_{k=0}^{\infty} a_k x^{k+1} = \sum_{k=1}^{\infty} a_{k-1} x^k$$

So the ODE is $y'' - xy = 0$

$$\Rightarrow \sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k - \left(\sum_{k=1}^{\infty} a_{k-1} x^k \right)$$

$$= 2a_2 + \sum_{k=1}^{\infty} (k+2)(k+1) a_{k+2} x^k - \sum_{k=1}^{\infty} a_{k-1} x^k$$

$$a_{3n+1} = \frac{\;}{(3)(4)(6)(7)\cdots(3n)(3n+1)}.$$

In other words, if we write down the series for y, it has two parts

$$\begin{aligned}
y &= \left(a_0 + \frac{a_0}{6}x^3 + \frac{a_0}{180}x^6 + \cdots + \frac{a_0}{(2)(3)(5)(6)\cdots(3n-1)(3n)}x^{3n} + \cdots\right) \\
&\quad + \left(a_1 x + \frac{a_1}{12}x^4 + \frac{a_1}{504}x^7 + \cdots + \frac{a_1}{(3)(4)(6)(7)\cdots(3n)(3n+1)}x^{3n+1} + \cdots\right) \\
&= a_0\left(1 + \frac{1}{6}x^3 + \frac{1}{180}x^6 + \cdots + \frac{1}{(2)(3)(5)(6)\cdots(3n-1)(3n)}x^{3n} + \cdots\right) \\
&\quad + a_1\left(x + \frac{1}{12}x^4 + \frac{1}{504}x^7 + \cdots + \frac{1}{(3)(4)(6)(7)\cdots(3n)(3n+1)}x^{3n+1} + \cdots\right).
\end{aligned}$$

We define

$$y_1(x) = 1 + \frac{1}{6}x^3 + \frac{1}{180}x^6 + \cdots + \frac{1}{(2)(3)(5)(6)\cdots(3n-1)(3n)}x^{3n} + \cdots,$$

$$y_2(x) = x + \frac{1}{12}x^4 + \frac{1}{504}x^7 + \cdots + \frac{1}{(3)(4)(6)(7)\cdots(3n)(3n+1)}x^{3n+1} + \cdots,$$

and write the general solution to the equation as $y(x) = a_0 y_1(x) + a_1 y_2(x)$. If we plug in $x = 0$ into the power series for y_1 and y_2, we find $y_1(0) = 1$ and $y_2(0) = 0$. Similarly, $y_1'(0) = 0$ and $y_2'(0) = 1$. Therefore $y = a_0 y_1 + a_1 y_2$ is a solution that satisfies the initial conditions $y(0) = a_0$ and $y'(0) = a_1$.

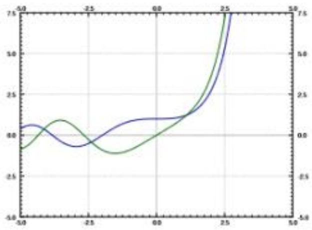

Figure: The two solutions y_1 and y_2 to Airy's equation.

See Figure above for the plot of the solutions y_1 and y_2. These functions have many interesting properties. For example, they are oscillatory for negative x (like solutions to $y'' + y = 0$) and for positive x they grow without bound (like solutions to $y'' - y = 0$).

Let us find a solution to the so-called *order n Hermite's equation*[2]:

$$y'' - 2xy' + 2ny = 0. \qquad P(x)=1,\ Q(x)=-2x,\ R(x)=2n$$

Let us find a solution around the point $x_0 = 0$. We try

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

We differentiate (as above) to obtain

$$y' = \sum_{k=1}^{\infty} k a_k x^{k-1},$$

$$y'' = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2}.$$

[2]Named after the French mathematician Charles Hermite (1822–1901).

Now we plug into the equation

$$\begin{aligned}
0 &= y'' - 2xy' + 2ny \\
&= \left(\sum_{k=2}^{\infty} k(k-1)a_k x^{k-2}\right) - 2x\left(\sum_{k=1}^{\infty} k a_k x^{k-1}\right) + 2n\left(\sum_{k=0}^{\infty} a_k x^k\right) \\
&= \left(\sum_{k=2}^{\infty} k(k-1)a_k x^{k-2}\right) - \left(\sum_{k=1}^{\infty} 2k a_k x^k\right) + \left(\sum_{k=0}^{\infty} 2n a_k x^k\right) \\
&= \left(2a_2 + \sum_{k=1}^{\infty}(k+2)(k+1)a_{k+2}x^k\right) - \left(\sum_{k=1}^{\infty} 2k a_k x^k\right) + \left(2na_0 + \sum_{k=1}^{\infty} 2n a_k x^k\right) \\
&= 2a_2 + 2na_0 + \sum_{k=1}^{\infty}\left((k+2)(k+1)a_{k+2} - 2ka_k + 2na_k\right)x^k.
\end{aligned}$$

reindex to match the power

As $y'' - 2xy' + 2ny = 0$, we have

$$(k+2)(k+1)a_{k+2} + (-2k+2n)a_k = 0, \qquad \text{or} \qquad a_{k+2} = \frac{(2k-2n)}{(k+2)(k+1)}a_k.$$

$$= 2a_2 + \sum_{k=1}^{\infty}(k+2)(k+1)a_{k+2}x^k - \sum_{k=1}^{\infty}a_{k-1}x^k$$

$$= 2a_2 + \sum_{k=1}^{\infty}\left((k+2)(k+1)a_{k+2} - a_{k-1}\right)x^k = 0$$

So $2a_2 = 0$

and $(k+2)(k+1)a_{k+2} - a_{k-1} = 0 \Rightarrow a_{k+2} = \dfrac{a_{k-1}}{(k+1)(k+2)}$.

$a_2 = 0 \Rightarrow a_5 = a_8 = a_{11} = \cdots = 0$

$a_0 \Rightarrow a_3 = \dfrac{a_0}{2\times 3}, \quad a_6 = \dfrac{a_3}{5\times 6} = \dfrac{a_0}{2\times 3\times 5\times 6}$

$a_9 = \dfrac{a_6}{8\times 9} = \dfrac{a_0}{2\times 3\times 5\times 6\times 8\times 9} =$

$a_1 \Rightarrow a_4 = \dfrac{a_1}{3\times 4}, \quad a_7 = \dfrac{a_4}{7\times 6} = \dfrac{a_1}{7\times 6\times 4\times 3} \cdots$

This recurrence relation actually includes $a_2 = -na_0$ (which comes about from the constant term $2a_2 + 2na_0 = 0$). Again a_0 and a_1 are arbitrary.

$$a_2 = \frac{-2n}{(2)(1)}a_0, \qquad a_3 = \frac{2(1-n)}{(3)(2)}a_1,$$
$$a_4 = \frac{2(2-n)}{(4)(3)}a_2 = \frac{2^2(2-n)(-n)}{(4)(3)(2)(1)}a_0,$$
$$a_5 = \frac{2(3-n)}{(5)(4)}a_3 = \frac{2^2(3-n)(1-n)}{(5)(4)(3)(2)}a_1, \quad \ldots$$

Let us separate the even and odd coefficients. We find that

$$a_{2m} = \frac{2^m(-n)(2-n)\cdots(2m-2-n)}{(2m)!},$$
$$a_{2m+1} = \frac{2^m(1-n)(3-n)\cdots(2m-1-n)}{(2m+1)!}.$$

Let us write down the two series, one with the even powers and one with the odd.

$$y_1(x) = 1 + \frac{2(-n)}{2!}x^2 + \frac{2^2(-n)(2-n)}{4!}x^4 + \frac{2^3(-n)(2-n)(4-n)}{6!}x^6 + \cdots,$$
$$y_2(x) = x + \frac{2(1-n)}{3!}x^3 + \frac{2^2(1-n)(3-n)}{5!}x^5 + \frac{2^3(1-n)(3-n)(5-n)}{7!}x^7 + \cdots.$$

We then write

$$y(x) = a_0 y_1(x) + a_1 y_2(x).$$

We remark that if n is a positive even integer, then $y_1(x)$ is a polynomial as all the coefficients in the series beyond degree n are zero. If n is a positive odd integer, then $y_2(x)$ is a polynomial. For example, if $n = 4$, then

$$y_1(x) = 1 + \frac{2(-4)}{2!}x^2 + \frac{2^2(-4)(2-4)}{4!}x^4 = 1 - 4x^2 + \frac{4}{3}x^4.$$

Singular points and the method of Frobenius

The behavior of ODEs at singular points can be complicated, for certain singular points, we can find a solution on at least one side of the singular point using a modification of the power series. Let us look at some examples before giving a general method.

Consider a simple first order equation

$$2xy' - y = 0.$$

$P(x) = 2x$, so at $x_0 = 0$, $P(x_0) = 0$, x_0 is a singular point.

Note that $x = 0$ is a singular point. If we try to plug in $y = \sum_{k=0}^{\infty} a_k x^k$, we obtain

$$0 = 2xy' - y = 2x\left(\sum_{k=1}^{\infty} k a_k x^{k-1}\right) - \left(\sum_{k=0}^{\infty} a_k x^k\right)$$
$$= a_0 + \sum_{k=1}^{\infty}(2ka_k - a_k)\,x^k.$$

The method for assuming (k is integer) $y = \sum_{k=0}^{\infty} a_k x^k$ doesn't work for the singular point. Need to use the method of Frobenius.

First, $a_0 = 0$. Next, the only way to solve $0 = 2ka_k - a_k = (2k-1)\,a_k$ for $k = 1, 2, 3, \ldots$ is for $a_k = 0$ for all k. Therefore, in this manner we only get the trivial solution $y = 0$.

Let us try $\boxed{y = x^r \text{ for some real number } r}$. Consequently our solution may only makes sense for positive x. Then $y' = rx^{r-1}$. So

$$0 = 2xy' - y = 2xrx^{r-1} - x^r = (2r-1)x^r.$$

Thus $r = \frac{1}{2}$ and so $y = x^{1/2}$. As the equation is linear, the general solution for positive x is

$$y = Cx^{1/2}.$$

If $C \neq 0$, then the derivative of the solution blows up at $x = 0$ (the singular point). There is only one solution that is differentiable at $x = 0$ and that's the trivial solution $y = 0$.

Assume $y = x^r$

then $y' = r \cdot x^{r-1}$

ODE becomes:

$2xy' - y = 2x \cdot rx^{r-1} - x^r$

$= 2rx^r - x^r = 0$

$\Rightarrow r = \frac{1}{2}$.

so $y = C \cdot x^{\frac{1}{2}}$

Not every problem with a singular point has a solution of the form $y = x^r$, of course. But perhaps we can combine the methods. What we will do is to try a solution of the form

For singular points : $y = x^r f(x) = x^r \sum_{k=0}^{\infty} a_k x^k$, k is integer, r is real number

for positive x, where $f(x)$ is a power series.

Consider the equation $P(x) = 4x^2 \quad Q(x) = -4x^2 \quad R(x) = 1-2x$

$$4x^2 y'' - 4x^2 y' + (1 - 2x)y = 0,$$ At $x_0 = 0$, $P(x_0) = 0$, so $x_0 = 0$ is a singular point.

and again note that $x = 0$ is a singular point.
Let us try

$$y = x^r \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} a_k x^{k+r},$$

where r is a real number, not necessarily an integer. Again if such a solution exists, it may only exist for positive x. First we find the derivatives

$$y' = \sum_{k=0}^{\infty} (k + r)\, a_k x^{k+r-1},$$

$$y'' = \sum_{k=0}^{\infty} (k + r)\,(k + r - 1)\, a_k x^{k+r-2}.$$

We plug those into our equation:

$$\begin{aligned}
0 &= 4x^2 y'' - 4x^2 y' + (1 - 2x)y \\
&= 4x^2 \left(\sum_{k=0}^{\infty} (k + r)\,(k + r - 1)\, a_k x^{k+r-2} \right) - 4x^2 \left(\sum_{k=0}^{\infty} (k + r)\, a_k x^{k+r-1} \right) \\
&\quad + (1 - 2x) \left(\sum_{k=0}^{\infty} a_k x^{k+r} \right)
\end{aligned}$$

expand and simplify

$$\begin{aligned}
&= \left(\sum_{k=0}^{\infty} 4(k + r)\,(k + r - 1)\, a_k x^{k+r} \right) \\
&\quad - \left(\sum_{k=0}^{\infty} 4(k + r)\, a_k x^{k+r+1} \right) + \left(\sum_{k=0}^{\infty} a_k x^{k+r} \right) - \left(\sum_{k=0}^{\infty} 2a_k x^{k+r+1} \right)
\end{aligned}$$

reindex to match powers

$$\begin{aligned}
&= \left(\sum_{k=0}^{\infty} 4(k + r)\,(k + r - 1)\, a_k x^{k+r} \right) \\
&\quad - \left(\sum_{k=1}^{\infty} 4(k + r - 1)\, a_{k-1} x^{k+r} \right) + \left(\sum_{k=0}^{\infty} a_k x^{k+r} \right) - \left(\sum_{k=1}^{\infty} 2a_{k-1} x^{k+r} \right) \\
&= 4r(r-1)a_0 x^r + a_0 x^r + \sum_{k=1}^{\infty} \left(4(k+r)(k+r-1)a_k - 4(k+r-1)a_{k-1} + a_k - 2a_{k-1} \right) x^{k+r} \\
&= \underbrace{\boxed{\left(4r(r - 1) + 1\right) a_0 x^r}}_{=0} + \sum_{k=1}^{\infty} \Big(\left(4(k + r)\,(k + r - 1) + 1\right) a_k - \left(4(k + r - 1) + 2\right) a_{k-1} \Big)\, x^{k+r}.
\end{aligned}$$

indical equation $\Rightarrow r = \frac{1}{2}$.

In order to have a solution we must have $\left(4r(r - 1) + 1\right) a_0 = 0$, we suppose $a_0 \neq 0$,

$$4r(r - 1) + 1 = 0.$$

This equation is called the *indicial equation*. This particular indicial equation has a double root at $r = \frac{1}{2}$.
If we plug in $r = \frac{1}{2}$ and solve for a_k, we get

$$a_k = \frac{4(k + \frac{1}{2} - 1) + 2}{4(k + \frac{1}{2})\,(k + \frac{1}{2} - 1) + 1}\, a_{k-1} = \frac{1}{k}\, a_{k-1}.$$

Let us set $a_0 = 1$. Then

$$a_1 = \frac{1}{1} a_0 = 1, \qquad a_2 = \frac{1}{2} a_1 = \frac{1}{2}, \qquad a_3 = \frac{1}{3} a_2 = \frac{1}{3 \cdot 2}, \qquad a_4 = \frac{1}{4} a_3 = \frac{1}{4 \cdot 3 \cdot 2}, \qquad \cdots$$

Extrapolating, we notice that

$$a_k = \frac{1}{k(k - 1)(k - 2) \cdots 3 \cdot 2} = \frac{1}{k!}.$$

In other words,

$$y = \sum_{k=0}^{\infty} a_k x^{k+r} = \sum_{k=0}^{\infty} \frac{1}{k!} x^{k+1/2} = x^{1/2} \sum_{k=0}^{\infty} \frac{1}{k!} x^k = x^{1/2} e^x.$$